

IONA 2.0 Project Proposal

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Abstract—The goal of this project is to integrate a real-time active visual feedback with the optimized design of the Intelligent Robotic Nursing Assistance (IONA) for teleoperation tasks. The optimization in this design includes a new chest that enhances arms manipulation and an active neck that enables a real-time immersive visual feedback to make the teleoperation task easier and more efficient.

Index Terms—teleoperation, active visual feedback, human-robot interaction

I. INTRODUCTION

This project exhibits an expanded and optimized design of a shared autonomous mobile humanoid robot used for remote nursing assistance. While the earlier design demonstrated a versatile supporting structure that allowed for flexible integration of various components, the new version introduced several improvements. Such improvements include an enhanced motorized chest that further improves the system's manipulation workspace. This incorporation can greatly enhance the robot's manipulability and reachability, especially in a cluttered spaces [1]. Additionally, an active neck will be integrated to provide real-time immersive visual feedback, allowing teleoperation to be more efficient and intuitive.

The primary goal of this expanded design is to optimize the Intelligent Robotic Nursing Assistance (IONA) system, developed by the Human-inspired Robotics (HiRo) Lab at Worcester Polytechnic Institute (WPI), to enable more efficient teleoperation tasks. These enhancements include improved manipulability and a more user-friendly interface that can control the robot via virtual reality (VR) devices, enabling the user to have real-time dynamic visual feedback as the robot's active neck mirrors the human's head movement. This new system aims to reduce the human's workload while also ensuring precise robotic manipulation in diverse caregiving settings.

The application of this advanced system is particularly timely, as nursing robots are expected to be deployed in various caregiving facilities such as hospitals and nursing homes in the near future. Such robots will help caregivers provide remote nursing and living assistance, reducing the risk of infections, while also helping to alleviate work stresses and discomfort [1]. Nursing assistance systems can also help to address the shortage of healthcare workers, while also supporting sustainability in healthcare.

II. RELATED WORK

A. Teleoperated Nursing Robots

1) *ABB's YuMi Robot*: G. Yang et al, [8] have studied the potential of using teleoperated robots to reduce instances of COVID-19 exposure among healthcare workers, especially given the personal protective equipment (PPE) shortages and the high risk of contracting the infection. A developed tele-operational and telesurgery system known as ABB's YuMi can be used to limit instances of infections by limiting physical contact with patients.



Fig. 1. ABB's YuMi Robot

The YuMi system is a telerobotic technology consisting of a dual-arm body that can be connected to a tablet device to enable communication between the operator and the robot. Operators are able to control the system using motion-capture suits to remotely control the robot, thereby limiting staff exposure when caring for patients with COVID-19. Some of the important functionalities of the YuMi robot include remote patient monitoring, medication delivery, instrument manipulation, and sterilization, essential routine tasks of a healthcare worker. Furthermore, such a system received positive feedback for its ability to reduce the exposure of healthcare workers to infectious diseases. [8]

The YuMi robot has also been adapted by Lyu et al., for the purpose of working with elderly dementia patients, by providing caregivers the remote assistance capabilities. Dementia care specifically often places a significant amount of physical and emotional stress on caregivers, therefore to alleviate these demands, Lyu and colleagues used the ABB's

YuMi dual-arm collaborative robot, to teleoperate through a wearable motion capture device [9].

This system facilitates an intuitive interaction for caregivers and operators reducing their physical strain. The telerobotic system operates by capturing the operator's hand movements through a motion capture suit, specifically the Perception Neuron suit, which is able to record hand position and orientation. This information is then transmitted to a computer running the Robot Operating System (ROS), in order to map the robot's movements in real-time. The robot is able to mirror the operator's hand motion with high precision, allowing it to complete routine tasks such as handling medication, retrieving items, and assisting patients. The accurate mapping of the hand motion enables teleoperated tasks to be performed, which is especially beneficial for elderly patients, as assistance is often required even when the caregiver is not there [9].

The system also included an interface with a live web camera that provided operators with visual feedback on the robot's actions, allowing for precise and real-time adjustments to be made when necessary. The visual feedback system is critical for task completion, or detailed manipulation, by enhancing the safety and effectiveness of the robot's movements [9].

Lyu et al., conducted experiments to track the accuracy of the robot when tasked with mimicking the operator's hand trajectory across different paths, such as lines, arcs and triangles. From this experiment, the YuMi robot exhibited high tracking accuracy with minimal positional errors, which emphasizes the system's potential to replicate complex hand movements. Demonstrations involving medication handling or remotely assisting an elderly woman further showed the system's usability. This teleoperated robot shows potential for enhancing dementia care, by easing the caregiver's burdens and physical strains by offering hands-free assistance, to support elderly patients at home remotely. However, despite its potential, the researchers acknowledge time delays with path mapping and the need to improve real-time responsiveness and dual-arm manipulation [9].

2) *Tele-Robotic Intelligent Nursing Assistant (TRINA)*: Teleoperated robotic systems have been employed for use to support healthcare workers and caregivers, in high-need environments, such as hospitals, care facilities, or patient homes. Yang et al., [7], introduced a Tele-Robotic Intelligent Nursing Assistant (TRINA) that would aid healthcare workers by minimizing their exposure to infectious diseases, such as COVID-19. TRINA consists of a mobile manipulator, a user-friendly console, and an integrated software that can automate or semi-automate tasks. Moreover, TRINA's teleoperation system enables human operators to control the robot through direct teleoperation, using game controllers, haptic feedback devices, or other interfaces. Additionally, the operator console offers visual feedback via multiple cameras, such as two RealSense 3D cameras which are mounted to the robot's wrists, or a 180° panoramic camera helping to construct a map of the environment to enhance situational awareness and efficiency at completing a task [7].

Through testing, TRINA showed to struggle with tasks

requiring fine dexterity such as opening packages or manipulating smaller components. Additionally, during routine nursing tasks such as preparing food and transporting medical devices, TRINA's manipulation success rate was high at 78 percent, however, the time to complete the tasks was often much longer than a human's. Yang and colleagues ultimately recommend enhancing TRINA's manipulation and improving the user interface to limit the learning curve for users. [7].

Previous research in the HiRo Laboratory at WPI explore how human operators adapt to the TRINA's interface multi-camera system described above. By analyzing how participants learned to manage the multi-camera system for tasks like object stacking, results highlight the users' preferences and adaptation to complex visual tasks, emphasizing the importance of intuitive, perception-based camera control. [6]

The study identifies human tendencies for natural head or gaze control in camera manipulation, which enhances depth perception and spatial awareness. These insights indicate that reducing the need for switching between cameras lowers cognitive load and aligns more closely with human visual behavior [6]. These results support our direction of work as it highlights the importance of a natural and intuitive camera control, and shows the potential for an active interface that mimics the operator's head movement, enabling an active real-time feedback.

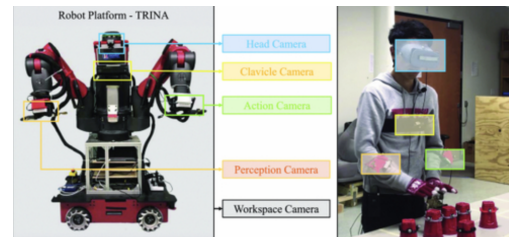


Fig. 2. TRINA humanoid robot multi-camera system (left) and camera set-up on the operator (right).

The study also revealed that, while performing the teleoperated tasks, the operators instinctively relied on tactile cues, such as tapping or aligning objects by touch, to supplement visual feedback. This finding shows that an integration of basic haptic feedback would aid in depth perception, improving task accuracy, without overwhelming cognitive capacities [6].

T.-C. Lin et al., [4], develops a hierarchical evaluation framework to evaluate three interface designs: a gamepad, a stylus, and a motion capture-based motion mapping; in order to assess performance, workload, and ease of learning. The study aims to identify which is the most ergonomic and intuitive design for the control of TRINA, that is low in physical and cognitive workload for non-technical users, such as nurses.

The first user study compared the interfaces' usability for 11 participants, including nursing students and registered nurses, using objective metrics, such as task completion time, errors and mode switches, and subjective metrics, such as NASA-TLX or System Usability Scale (SUS).

The results indicated that motion mapping was the most intuitive and efficient design, as it enables simultaneous control of the robot's arms and base, increasing efficiency and reducing cognitive workload. It also is the most intuitive, even for those without prior experience, due to its nature of mirroring whole-body movements. Despite the intuitiveness and efficiency, mirroring physical movements directly resulted in significant muscle fatigue [4].

The second user study introduced electromyography (EMG) sensors to measure muscle fatigue in 14 muscle groups, aiming to quantify the physical demands of using the motion mapping interface. The study showed that tasks like steady wrist camera control and object alignment induced significant fatigue, suggesting the need for adaptive ergonomic support. Because of these results, to mitigate physical fatigue, the authors proposed shared autonomy features, where the robot assists with specific high-fatigue tasks, such as object grasping, by automating some movements. This approach reduced physical demands while maintaining task performance and accuracy [4]. These insights show the potential of the integration of IONA with an active visual feedback as it would enhance an intuitive and efficient interface for teleoperated tasks, while also including shared autonomy features for potential high-fatigue tasks.

B. IONA 1.0

The most recent work completed by Boguslavskii et al., [1] implemented the latest version of the Intelligent Robotic Nursing Assistance (IONA) focusing on the integration of shared autonomy methods to enhance the robot's manipulation capabilities and reduce the human effort in coordinating both the motorized chest and manipulator arms. In this case, shared autonomy supports collision avoidance, base-arm coordination, and controlled manipulation. The researchers also proposed a fatigue-adaptive shared autonomy feature, which enables human-robot collaboration (HRC), by having the robot autonomously handle precise, often fatigue-inducing actions such as grasping; while the human guides the physical movement of the robot [1]. Despite IONA's potential and capabilities, shared autonomy is not the scope of this project.

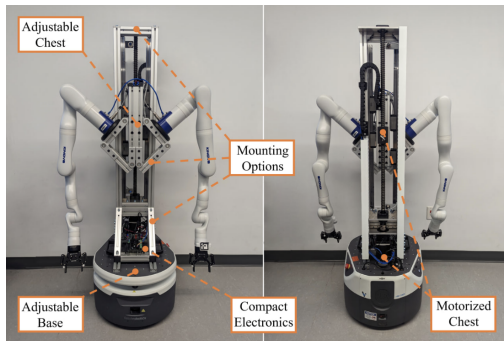


Fig. 3. IONA Robot

Systems such as MetaQuest and KAT Walk VR Treadmill, have shown the benefits of using a VR interface for control, as

the operator is able to guide the robot's movements. This proposed system allows for real-time, ego-centric visual feedback to enhance the robot's ability to complete tasks that require high precision. The graphical user interface incorporated in the system facilitates monitoring of the robot's states, camera viewpoints, and control modes for real-time tracking of the robot's location and navigation plans. Moreover, a shared control approach reduces the need for human intervention and can optimize robot performance, making it suitable for cluttered and unstructured environments [1].

The software architecture used for the mobile nursing robot developed by Boguslavskii et al. integrates a Linux-based system running the Robot Operating System (ROS). ROS controls and manages various components, including the manipulator arm, mobile base, cameras, and motorized chest. The operator console utilizes Unity on Windows, supporting multiple control interfaces from VR systems, while UnityRobotics-Hub TCP/IP is employed for video streaming and command transmission to the robot [1].

The mobile nursing assistance system also incorporates a Relaxed Inverse Kinematics system (IK), described by Rakita et al., to resolve the inverse kinematics for the manipulators, to ensure smooth motion, while avoiding joint-space discontinuities [5].

C. Open TeleVision

Cheng et al., [3], introduce an Open-TeleVision framework that includes several components that can significantly enhance the development of a teleoperation system for a remote nursing assistance robot. Using VR devices along with the enabled active visual feedback, this system can offer an immersive interface for controlling the robot. This system streams data such as the operator's hand, head, and wrist poses from the VR device to a web server based on Vuer. This server then handles the human-to-robot motion mapping, to ensure precise control of the robot (PyPi) [2], while the robot streams stereo video back to the operator, completing the communication loop. This system enables remote teleoperation by linking the VR device with the robot's hardware over the internet. This is particularly valuable for achieving precise teleoperation tasks in healthcare settings, as the robot would be dealing with delicate patient care and object manipulation [3].

A gimbal is included which accounts for two degrees of freedom (DoFs), that plays the role of an active neck, enabling the camera to move in sync with the operator's head, providing a more natural and effective way of controlling the nursing robot in cluttered environments. The integration of the active neck with a stereo RGB camera, specifically a ZED Mini stereo camera, provides real-time stereo RGB streaming that enhances a better spatial understanding during teleoperation and improves the robot's ability to move its head in coordination with the operator. The increased spatial awareness enabled by the stereoscopic video allows for better control of the robot in unstructured environments, where even small obstacles might obstruct the robot's path and interfere with the completion of a task. This aspect of the framework

could improve the performance of teleoperated nursing robots, by ensuring more accurate interactions with patients, reducing chances of error [3].

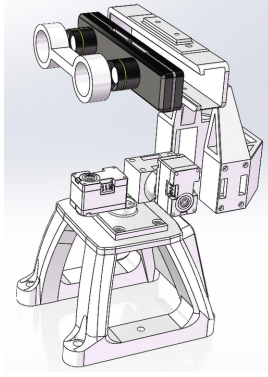


Fig. 4. Active neck components for active visual feedback

The system's ability to remap human motions demonstrates that nurses and caregivers can effectively guide the robot through various tasks while exerting minimal effort. Real-time views ensure that the operator maintains sight and focus on key areas throughout complex tasks, such as handling medical equipment or helping a patient with mobility. As well, real-time feedback is crucial for ensuring smooth interactions between the operator and the robot. The ability to receive immediate visual updates from the robot, allows healthcare workers to react to such changes, with high accuracy and attention.

Although outside the domain of this project, Open-TeleVision also explored imitation learning, in which the robot actively learns and imitates the movements of the human head. This concept could be extended to nursing robots, by training these robots to learn from the movements of experienced nurses, enabling these robots to better navigate patient care, and improve task efficiency, by minimizing the need for constant human intervention [3]. Overall, the integration of VR-based active visual feedback into a nurse assistance system can assist in both operational accuracy and user-friendliness of the system for healthcare providers to use.

III. PROPOSED METHODS

We propose an integration of the inherent code of the Open-TeleVision framework proposed by Cheng et al. and the optimized design of the Intelligent Robotic Nursing Assistance (IONA) implemented by Boguslavskii et al. The main goal of the project is to successfully integrate IONA's teleoperation software architecture with the significant code in the Open-TeleVision framework that enables the active visual feedback. The assembly of the expanded IONA system, with the new chest design and active neck, will be essential as we won't be able to test the system without the physical structure, however, the main goal of the project is to properly integrate the software components. The overall process can be divided in three phases: (i) manufacturing phase, (ii) assembly phase and, (iii) software integration phase.

The manufacturing phase consists of the manufacturing of all the different and necessary components of IONA, shown in Figure 5, along with 3D-printing the components of the active neck shown in Figure 4.

The assembly phase consists of assembling from scratch IONA's optimized design, along with assembling the active neck, and finally integrate them in the same system.

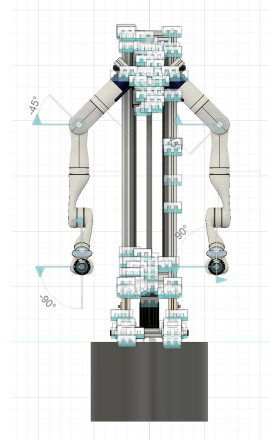


Fig. 5. Optimized Y chest design for the nurse assistance system.

Once we will have the base system of IONA with the teleoperation functionality, we will be ready to delve into the software integration phase, which will be the most complex and time-consuming to develop.

In order to successfully integrate these two systems, we will need, at first, to develop a web server on Vuer in order to stream the real-time visual feedback from the ZED Mini camera to the VR device used by the human operator, and to stream the human's movement data from the VR device to the server. After having developed the web application, we will integrate the inherent code of the Open-TeleVision framework with the IONA software architecture in order to map the operator's head movements to IONA's reference frame. Furthermore, an lower level code integration will be needed in order to be properly communicating and working with the different devices. A diagram of components highlighting the most important features of the Open-TeleVision system implemented by Cheng et al. is shown in Figure 6.

IV. EXPERIMENT

In order to evaluate the integration of active visual feedback with IONA, it is necessary to focus on three main questions to help us guide our research: how active visual feedback improves (i) task performance, (ii) interface and interaction intuitiveness and transparency, along with (iii) overall human's mental workload.

To properly assess the potential of our systems integration, we will refer to the hierarchical framework for human-robot interface evaluation proposed by T.-C. Lin et al., [4], shown below in Figure 7.

To truly respond to these three questions we need to carry on a comparative user study that compares the users' experiences

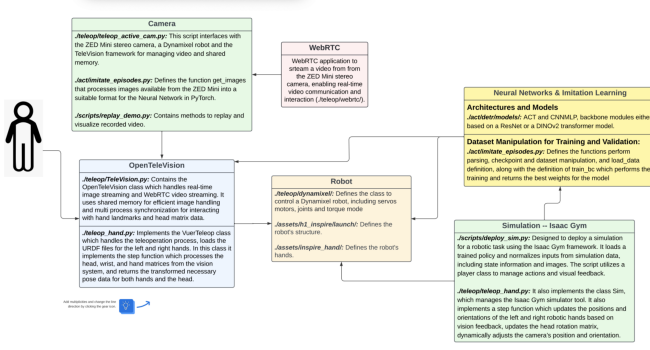


Fig. 6. Open-Television system Diagram of Components.

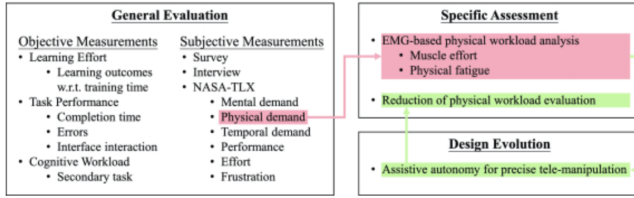


Fig. 7. A proposed hierarchical framework for the evaluation and evolution of human-robot interface.

in using the old and the new version of IONA. To evaluate the users' experience we will use the proposed hierarchical framework, using both objective and subjective measurements.

Among the objective metrics, to evaluate the intuitiveness and transparency of the interface, we will need to focus on the learning effort and curve, which can be assessed using learning outcomes, such as task completion time, error and success rate before and after being introduced to the interface. We could also evaluate the interface by asking participants to perform the teleoperated tasks several times and, by looking at the difference in tasks' completion times using both robots, we can understand if the integration of the active visual feedback improves the learning curve and makes the teleoperation tasks easier.

To properly evaluate how the task performance is influenced by the systems integration, we will use objective measurements such as completion time, success and error rate, along with the overall user-interface interaction. The user performs several basic manipulation tasks such as pick-and-place, sorting objects, inserting, unloading and folding. By comparing these task performance values, we can get a better understanding on how active visual feedback really influence the execution of a task.

For the workload evaluation, we are going to use both objective and subjective metrics. Regarding the objective measures, we will carry on an experiment previously performed by T.-C. Lin et al., [4], where, while the participants are teleoperating IONA, we will ask the users to solve simple arithmetic questions. T.-C. Lin et al., explain that, based on the time that the user takes to solve the problem, we can compare how mentally demanding IONA's teleoperation is,

whether with dynamic or static visual feedback.

For our subjective metrics, we plan to use the NASA-Task Load Index (NASA-TLX) survey to record the user's self-evaluation of mental and physical workload, temporal demands, performance, effort, and frustrations [4], and we also plan to use the System Usability Scale to get a reliable system usability score, covering aspects like complexity, comfort, and confidence in using the system. Furthermore, if time allows, we plan to interview participants at the end of the study to get further insights on the interface and to understand directions for improvements.

By using the proposed framework in our comparative user study we can really understand the benefits and limitations of real-time dynamic visual feedback.

V. PRELIMINARY RESULTS

At this time we are still in the manufacturing and research phase. In the past couple of weeks I delved into the OpenTelevision's software and I started identifying the core functions and methods used to stream the live video from the ZED Mini camera to the VR device through the web server. I've been going through how to implement the web server to connect the VR device and the robot's system, along with mapping the joints values from the VR device to the Robot's active neck Dynamixel Motors. Also, since the IONA system does not support deep learning features associated to the ZED SDK for image manipulation such as object detection, tracking and image segmentation, we will need to purchase and integrate a NVIDIA GPU. After performing a comparative study of different NVIDIA GPUs, we concluded that for mobility and power efficiency, the Jetson Xavier is recommended. It balances processing power, stereo video support, and GPU acceleration for real-time depth sensing.

VI. SCHEDULE

Phase	Sub-Tasks	Task Division	Deadline
Manufacturing	Manufacture IONA's parts	Boguslavskii	11/3/2024
	3D-Print Active Neck parts	Marcantoni	
Assembly	IONA Assembly	Boguslavskii assisted by Marcantoni	11/15/2024
	Active Neck Assembly	Marcantoni	
Software Integration	Identify significant methods and files from Open-Television code.	Marcantoni	11/25/2024
	Web server development on Vuer	Marcantoni	
	Check the real-time connection in between server, camera and VR device.	Boguslavskii & Marcantoni	
Evaluation	Integrate inherent code from the Open-Television framework to map the operator's head movements into IONA's work frame.	Boguslavskii & Marcantoni	12/10/2024
	Objective and subjective recordings during experiments	Boguslavskii & Marcantoni	
	Recording analysis	Boguslavskii & Marcantoni	

Fig. 8. Project's Timeline and Task Division.

The table shown above in Figure 8 indicates the different phases of the integration and evaluation process, the necessary subtasks to complete each phase, the task division and finally the set deadlines for the completion of each phase.

REFERENCES

- [1] N. Boguslavskii, Z. Zhong, L. M. Genua, and Z. Li, "A Shared Autonomous Nursing Robot Assistant with Dynamic Workspace for Versatile Mobile Manipulation," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 7040–7045, Oct. 2023, doi: <https://doi.org/10.1109/iros55552.2023.10342401>.
- [2] "vuer," PyPI, Oct. 20, 2024. <https://pypi.org/project/vuer/> (accessed Oct. 20, 2024).

- [3] X. Cheng, J. Li, S. Yang, G. Yang, X. Wang, and U. San, "Open-TeleVision: Teleoperation with Immersive Active Visual Feedback." Accessed: Oct. 20, 2024. [Online]. Available: <https://robot-tv.github.io/resources/television.pdf>
- [4] T.-C. Lin, A. U. Krishnan, and Z. Li, "Intuitive, efficient and ergonomic tele-nursing robot interfaces: Design evaluation and evolution," *Transactions on Human-Robot Interaction*, vol. 10, no. 2, pp. 89-108, 2021.
- [5] D. Rakita, B. Mutlu, and M. Gleicher, "Relaxed IK: Real-time synthesis of accurate and feasible robot arm motion," in *Robotics: Science and Systems*, Pittsburgh, PA, USA, 2018, pp. 26-30.
- [6] T.-C. Lin, A. U. Krishnan, and Z. Li, "Perception-Motion Coupling in Active Telepresence: Human Behavior and Teleoperation Interface Design," *J. Hum.-Robot Interact.* 12, 3, Article 31 (September 2023), 24 pages. <https://doi.org/10.1145/3571599>
- [7] Z. Li, P. Moran, Q. Dong, R. J. Shaw and K. Hauser, "Development of a tele-nursing mobile manipulator for remote caregiving in quarantine areas," 2017 IEEE International Conference on Robotics and Automation (ICRA), Singapore, 2017, pp. 3581-3586, doi: 10.1109/ICRA.2017.7989411.
- [8] G. Yang, H. Lv, Z. Zhang, et al. "Keep Healthcare Workers Safe: Application of Teleoperated Robot in Isolation Ward for COVID-19 Prevention and Control," *Chin. J. Mech. Eng.* 33, 47 (2020). <https://doi.org/10.1186/s10033-020-00464-0>
- [9] H. Lyu, G. Yang, H. Zhou, X. Huang, H. Yang and Z. Pang, "Teleoperation of Collaborative Robot for Remote Dementia Care in Home Environments," in *IEEE Journal of Translational Engineering in Health and Medicine*, vol. 8, pp. 1-10, 2020, Art no. 1400510, doi: 10.1109/JTEHM.2020.3002384.